cfdpost: main formulas and references

Nodal gradients

The cell gradient is computed with Green–Gauss [3, 4]. Quadrilateral faces are split into four triangles t around the face centroid:

$$\nabla f|_c = \frac{1}{V_c} \sum_{t \in \partial c} \bar{f}_t \, \boldsymbol{S}_t, \qquad V_c = \frac{1}{3} \sum_{t \in \partial c} \bar{\boldsymbol{x}}_t \cdot \boldsymbol{S}_t.$$

This is exact for linear fields on tetrahedra, pyramids, prisms and hexahedra. Cell gradients are averaged to the nodes with volume weights:

$$\nabla f_i = \frac{\sum_{c \ni i} |V_c| \, \nabla f|_c}{\sum_{c \ni i} |V_c|}.$$

The dual volume and mesh size are $V_i = \sum_{c \ni i} |V_c|/n_c$ and $h_i = V_i^{1/d}$.

Iso-surfaces, slices and cuts

Each face whose node signs change contributes the directed segment from its "down" crossing ($+ \to -$) to its "up" crossing ($- \to +$); a cell's segments chain into the output polygons. This is related to marching cubes and marching tetrahedra [5, 12], but needs no tetrahedral subdivision [6]. The crossing point on edge (a, b) is

$$\boldsymbol{x} = \boldsymbol{x}_a + t(\boldsymbol{x}_b - \boldsymbol{x}_a), \qquad t = \frac{\phi^* - \phi_a}{\phi_b - \phi_a},$$

and fields are interpolated with the same t. Ambiguous faces are resolved by the sign of the face-centre value [16]. A slice is the iso-surface of $\phi = (\boldsymbol{x} - \boldsymbol{x}_0) \cdot \hat{\boldsymbol{n}}$.

Shock detection

Lovely–Haimes [13, 14]. The shock surface is $M_n = M_n^*$ (default 1), with

$$M_n = \frac{\boldsymbol{V}}{a} \cdot \frac{\nabla p}{|\nabla p|}, \qquad a = \sqrt{\gamma p / \rho}.$$

Polygons are kept if

$$|\nabla p| > k \, \mathrm{median} \, |\nabla p| \quad \text{and} \quad \mu_t / \mu < r_{\max}.$$

The presets $(M_n^*, k, r_{\max})$ are: default $(1, 3, 10^3)$, inclusive $(1, 1.5, 10^3)$ and conservative $(1.05, 5, 10^2)$.

Ducros sensor [7], with $\theta = \nabla \cdot \boldsymbol{V}$:

$$\Phi = \frac{\theta^2}{\theta^2 + |\nabla \times \boldsymbol{V}|^2 + \varepsilon} \quad (\theta < 0), \qquad \Phi = 0 \text{ otherwise.}$$

Normalised pressure jump $\sigma_p = |\nabla p| \, h/p$, in the spirit of the pressure switch of [10].

Numerical schlieren $|\nabla \rho|$ [19].

Derived quantities

From [1]:

$$p_0 = p\Big(1 + \tfrac{\gamma-1}{2} M^2\Big)^{\frac{\gamma}{\gamma-1}}, \quad T_0 = T\Big(1 + \tfrac{\gamma-1}{2} M^2\Big), \quad s/c_v = \ln \frac{p}{\rho^\gamma}.$$

With $G = \nabla \boldsymbol{V}$, $S = \frac{1}{2}(G + G^T)$ and $\Omega = \frac{1}{2}(G - G^T)$:

$$Q = \tfrac{1}{2}\big(\|\Omega\|^2 - \|S\|^2\big) \text{ [9]}, \qquad \lambda_2 = \lambda_{\text{mid}}(S^2 + \Omega^2) \text{ [11]}.$$

The wind axes follow SU2 [8]:

$$\hat{\boldsymbol{e}}_D = (\cos\alpha\cos\beta, \sin\beta, \sin\alpha\cos\beta), \qquad \hat{\boldsymbol{e}}_L = (-\sin\alpha, 0, \cos\alpha).$$

Surface loads

Per unit dynamic pressure, with $\hat{\boldsymbol{n}}$ the unit normal pointing into the fluid:

$$\mathrm{d}\boldsymbol{F}/q_\infty = (-C_p\hat{\boldsymbol{n}} + \boldsymbol{C}_f)\,\mathrm{d}A.$$

C_p and $\boldsymbol{C}_f$ are linear on each triangle, and $\int \varphi_i\varphi_j\,\mathrm{d}A = \frac{A}{12}(1 + \delta_{ij})$ [20]. Hence

$$\boldsymbol{F}_p = -\tfrac{1}{3}\Big(\sum_i C_{p,i}\Big)\boldsymbol{S}, \qquad \boldsymbol{F}_v = \tfrac{A}{3}\sum_i \boldsymbol{C}_{f,i}, \qquad \boldsymbol{M} = \frac{A}{12}\sum_{i,j}(1 + \delta_{ij})(\boldsymbol{x}_i - \boldsymbol{x}_{\text{ref}}) \times (-C_{p,j}\hat{\boldsymbol{n}} + \boldsymbol{C}_{f,j}).$$

The coefficients are $C_L = \boldsymbol{F}\cdot\hat{\boldsymbol{e}}_L/S_{\text{ref}}$ and $C_M = \boldsymbol{M}/(S_{\text{ref}}L_{\text{ref}})$. Face normals use Newell's formula $\boldsymbol{N} = \sum_k \boldsymbol{x}_k \times \boldsymbol{x}_{k+1}$ [18]. A closed body satisfies $\oint x\,\hat{\boldsymbol{n}}\,\mathrm{d}A = V\hat{\boldsymbol{e}}_x$ [15].

Span loading

Triangles are clipped exactly at the strip planes [17]. In each strip,

$$\ell_k = c\,c_l = \frac{\boldsymbol{F}_k\cdot\hat{\boldsymbol{e}}_L}{\Delta\eta_k}, \qquad c_{l,k} = \frac{\ell_k}{c_k}.$$

The elliptic reference carries the same load [2]:

$$\ell_{\text{ell}} = \ell_0\sqrt{1 - (\eta/s)^2}, \qquad \ell_0 = \frac{\sum_k \ell_k\Delta\eta_k}{\sum_k \sqrt{1 - (\eta_k/s)^2}\,\Delta\eta_k}.$$

A strip is open (the markers do not close the section) when

$$\frac{|A^+ - A^-|}{\max(A^\pm)} > \tau, \qquad A^\pm = \tfrac{1}{2}\Big(\sum|S_z| \pm \sum S_z\Big).$$

The free-stream angles are taken from the mean far-field velocity: $\alpha = \operatorname{atan2}(U_z, U_x)$ and $\beta = \arcsin(U_y/|\boldsymbol{U}|)$.

Sections

On a cut loop, the trailing edge is the point furthest along the chord direction, and the leading edge is the point furthest from the trailing edge. With $\Delta\boldsymbol{x} = \boldsymbol{x}_{TE} - \boldsymbol{x}_{LE}$,

$$c = |\Delta\boldsymbol{x}|, \qquad \theta = \operatorname{atan2}(-\Delta x_z, \Delta x_x).$$

Same-topology meshes

Nodes are matched by global index:

$$\boldsymbol{d}_i = \boldsymbol{x}_i^{\text{def}} - \boldsymbol{x}_i^{\text{base}}, \qquad d_{n,i} = \boldsymbol{d}_i\cdot\hat{\boldsymbol{n}}_i, \qquad \Delta q_i = q_i^{\text{opt}} - q_i^{\text{base}},$$

where $\hat{\boldsymbol{n}}_i$ is the normalised sum of the area vectors of the faces around node i.

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